$$
\begin{cases}
\partial_t f & = \begin{array}{l} \sigma_x \Delta_x f - \mathrm{Pe} \nabla_x \cdot (v(\theta) f) \\ + \sigma_\theta \partial_\theta^2 f - \chi \partial_\theta (n(\theta) \cdot (\nabla K * \rho + \tau \nabla^2 K * \rho v(\theta)) f) \end{array} \\
f(t = 0) & = f_0
\end{cases}
$$

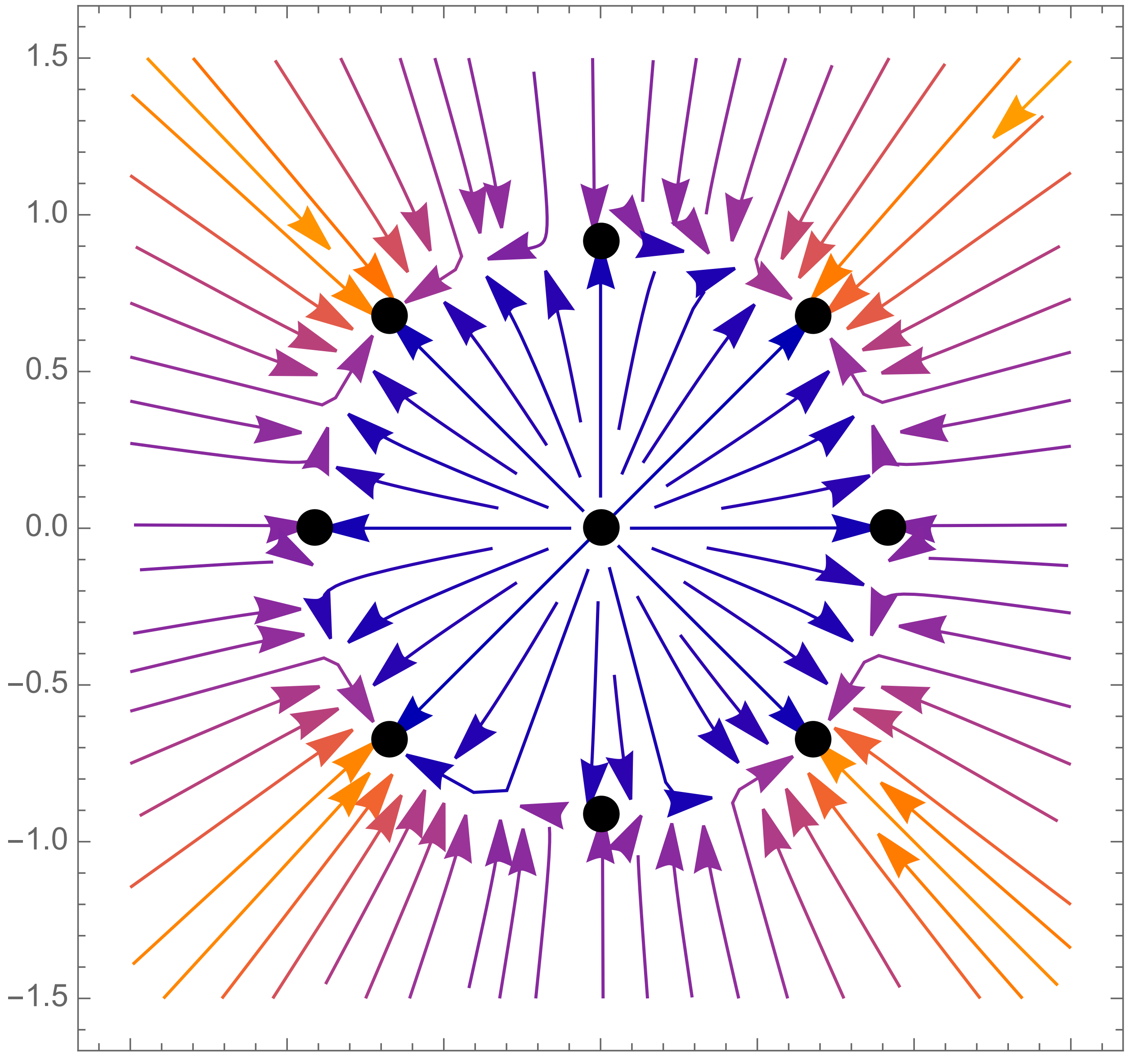
- What happens as $t \to \infty$?

- Part 2 (Rakotomalala & dW, in review, 2026):

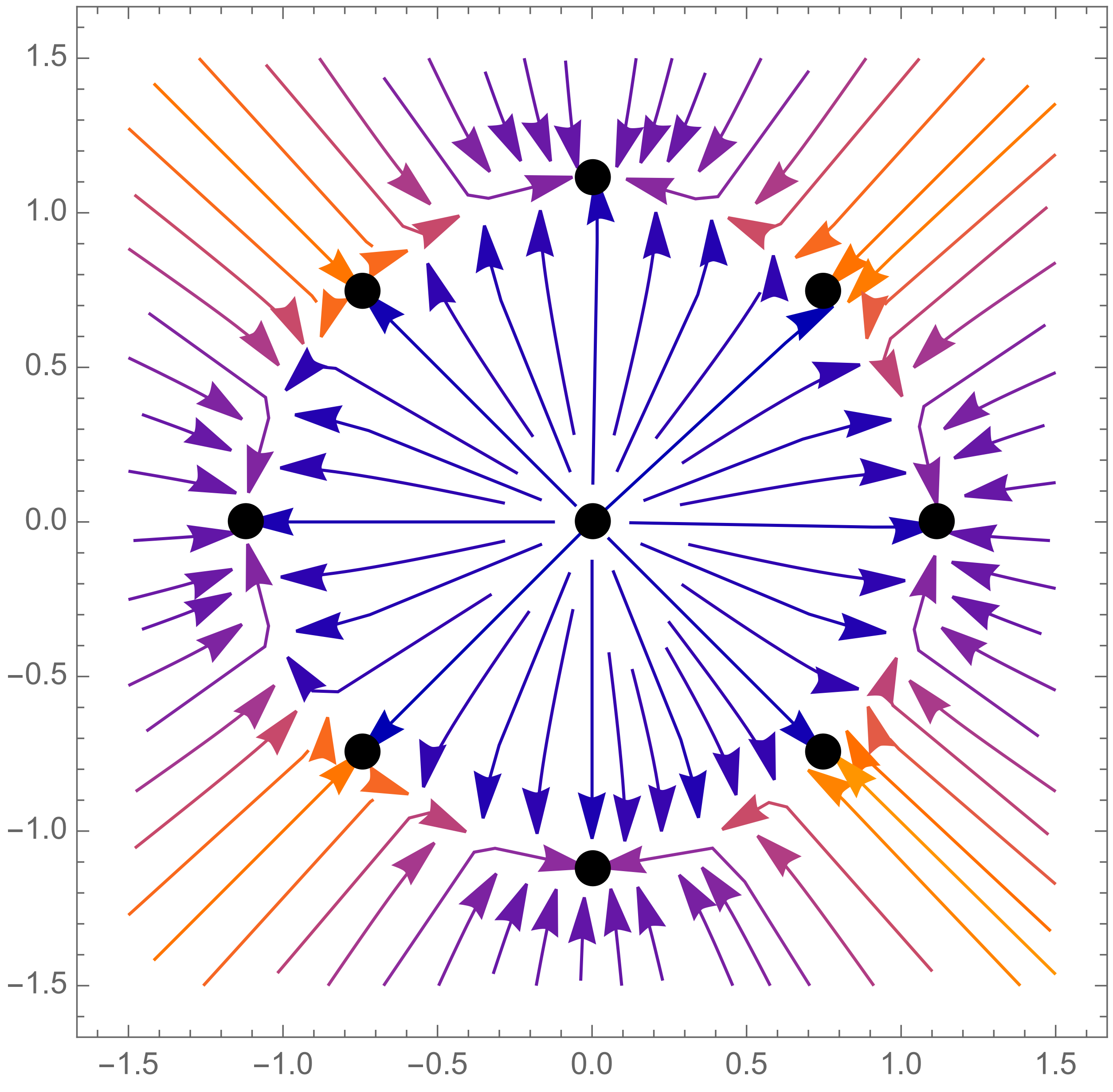
- Existence of two families of non-trivial stationary states

- Existence of critical $\tau = \tau^*$ super- to subcritical

- Lyapunov-Schmidt method: PDE $\Longleftrightarrow$ ODE



$$\tau < \tau^*$$



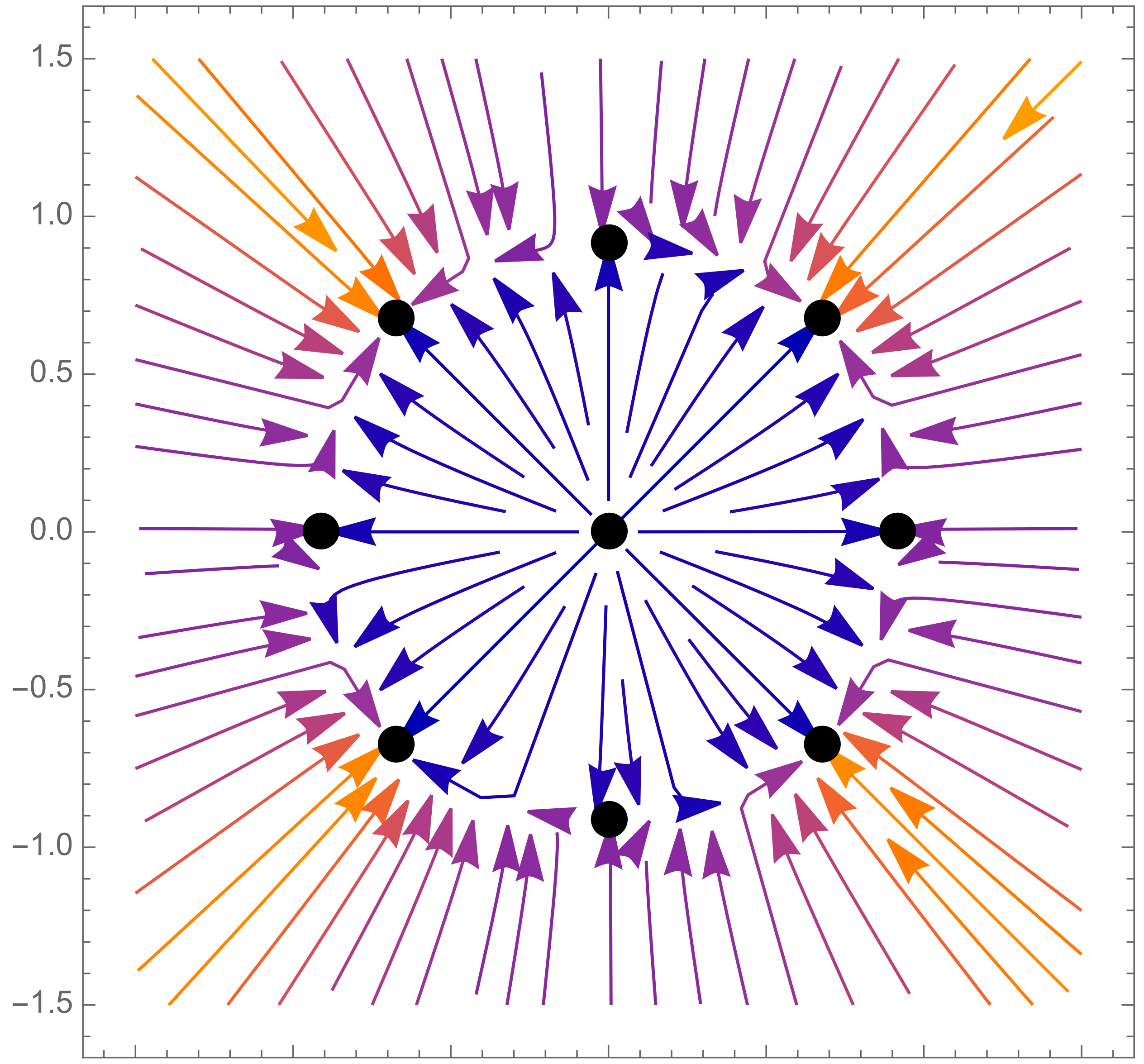
$$\tau > \tau^*$$

Ant dynamics

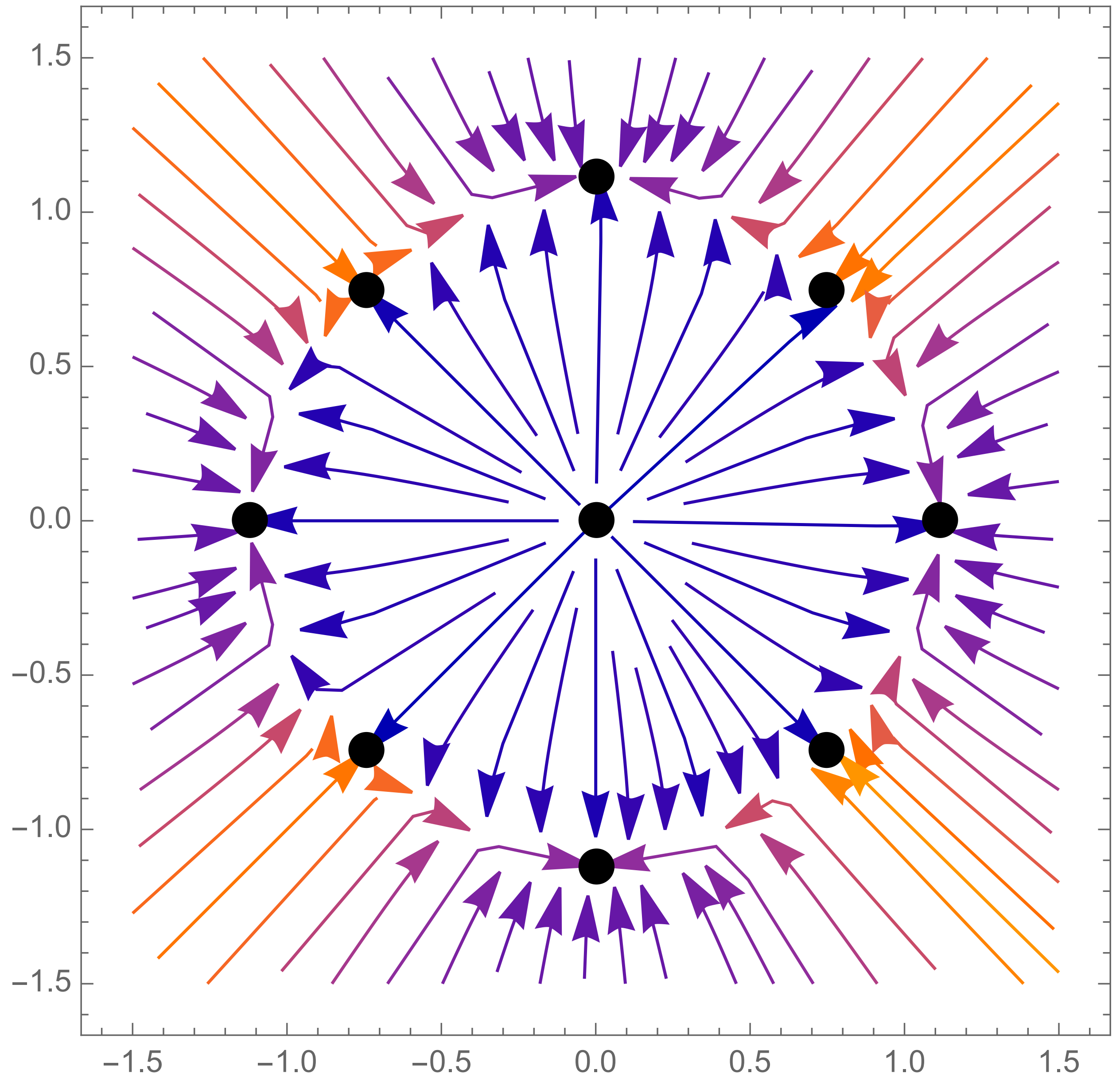
Ant dynamics

$$\begin{cases} \partial_t f & = \begin{array}{l} \sigma_x \Delta_x f - \mathrm{Pe} \nabla_x \cdot (v(\theta) f) \\ + \sigma_\theta \partial_\theta^2 f - \chi \partial_\theta (n(\theta) \cdot (\nabla K * \rho + \tau \nabla^2 K * \rho v(\theta)) f) \end{array} \\ f(t=0) & = f_0 \end{cases}$$

- What happens as $t \to \infty$?
- Part 2 (Rakotomalala & dW, in review, 2026):
  - Existence of two families of non-trivial stationary states
  - Existence of critical $\tau = \tau^*$ super- to subcritical
  - Lyapunov-Schmidt method: PDE $\Longleftrightarrow$ ODE



$\tau < \tau^*$



$\tau > \tau^*$

Ant dynamics

$$\begin{cases} \partial_t f & = \sigma_x \Delta_x f - \mathrm{Pe} \nabla_x \cdot (v(\theta) f) + \sigma_\theta \partial_\theta^2 f - \chi \partial_\theta (n(\theta) \cdot (\nabla K * \rho + \tau \nabla^2 K * \rho v(\theta)) f) \\ f(t=0) & = f_0 \end{cases}$$

Lyapunov-Schmidt method